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Misperceived Social Norms and Willingness to Act Against Climate Change

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Appendix A provides additional figures and tables. Appendix B contains the main survey instructions. Appendix C explains how the measures of economic preferences and moral universalism are derived. Appendix D presents information on the pre-analysis plan and deviations from it.

Appendices

Appendix A Supplementary Analyses

Table A.1: Comparison of the sample to the US population

Variable	Wave 1	Wave 2	US population (2019)
Female	51%	51%	51%
Age: 18-34	30%	30%	30%
Age: 35-54	32%	32%	32%
Age: 55+	38%	38%	38%
Education: Bachelor's degree or above	31%	31%	31%
Region: Northeast	17%	17%	17%
Region: Midwest	21%	21%	21%
Region: South	38%	38%	38%
Region: West	24%	24%	24%
Democrat*	54%	53%	52%
Republican*	46%	47%	47%
Income: Below 50k*	41%	45%	37%
Income: 50k to 100k*	39%	35%	31%
Income: Above 100k*	19%	20%	31%

Notes: Columns 1 and 2 display the summary statistics for the survey samples of waves 1 and 2, respectively. Column 3 displays summary statistics based on the American Community Survey 2019, with the following exceptions: The share of voters who lean towards the Democratic (Republican) Party is obtained from the annual totals of the Pew Research Center survey data in 2019. Variables marked with a (*) were not targeted by the quota-based sampling approach.

Table A.2: Relationship between donations and self-reported concrete climate behaviors

	Dependent variable: Donation (\$)					
	(1)	(2)	(3)	(4)	(5)	(6)
Restrict meat consumption	63.603*** (6.465)					
Avoid taking flights		74.641*** (6.520)				
Use car alternatives			66.675*** (6.440)			
Use electricity from green sources				13.785* (7.329)		
Adapt shopping behavior					78.708*** (6.334)	
Political engagement						76.922*** (6.343)
Constant	199.819*** (4.296)	181.223*** (5.060)	191.127*** (4.861)	222.518*** (3.821)	190.184*** (4.428)	193.612*** (4.349)
N	1,991	1,989	1,991	1,991	1,990	1,989

Notes: This table shows OLS regression estimates using respondents from wave 1 where the dependent variable is the donation decision (in \$). Each column regresses donations on a binary indicator taking value one for respondents who said that they engage in the concrete climate-friendly behavior, and zero otherwise.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.3: Validation of self-reported willingness to pay

	Dependent variable:						
	(1) Donation (in \$)	(2) Restrict meat consumption	(3) Avoid taking flights	(4) Use car alternatives	(5) Renewable energy use	(6) Adapt shopping behavior	(7) Politically support fight against GW
WTP 1%	81.340*** (6.265)	0.333*** (0.021)	0.207*** (0.021)	0.387*** (0.021)	0.283*** (0.018)	0.365*** (0.021)	0.442*** (0.020)
Constant	188.849*** (4.543)	0.252*** (0.013)	0.501*** (0.015)	0.342*** (0.014)	0.084*** (0.008)	0.284*** (0.014)	0.213*** (0.012)
N	1,993	1,998	1,996	1,998	1,998	1,997	1,996
R ²	0.076	0.114	0.044	0.148	0.120	0.133	0.200

Notes: This table shows OLS regression estimates using respondents from wave 1. “WTP 1%” is a binary indicator taking value one for respondents who are willing to give up 1% of their income for the fight against climate change. “Donation” is the amount donated that a respondent chose to donate to a pro-climate charity (incentivized). The dependent variables in columns 2–7 are binary indicators taking value one for respondents who reported to engage in the behavior indicated in the column header.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.4: Education and individual willingness to fight global warming

	Outcome: Donation (\$)	
	(1)	(2)
Democrat x college degree	-9.644 (7.618)	-8.857 (7.598)
Republican x college degree	-22.898** (8.971)	-22.496** (9.041)
N	1,975	1,975
Control group mean	225.21	225.21
Demographic controls	Yes	Yes
Preferences and moral universalism	Yes	Yes
Normative belief	Behavior belief	Norms belief

Notes: This table shows OLS regression estimates where the dependent variable is the donation decision (in \$) using respondents from wave 1. All regressions specifications are identical to those in Table 1, including demographic controls, economic preferences, moral universalism as well as normative beliefs as covariates. However, we replaced the “College degree” indicator with a “Democrat x college degree” and a “Republican x college degree” indicator.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.5: Determinants of norm misperceptions

	Dependent variable: Absolute prediction error (in percentage points)			
	Behavior belief		Norms belief	
	(1) Full sample	(2) Underestimators only	(3) Full sample	(4) Underestimators only
Democrat	-1.977*** (0.664)	-2.086** (0.867)	-3.215*** (0.815)	-3.397*** (0.944)
Age	0.059*** (0.021)	0.075*** (0.027)	0.129*** (0.026)	0.133*** (0.029)
Female	0.832 (0.666)	0.773 (0.867)	1.264 (0.807)	1.187 (0.932)
Log income	-0.657 (0.427)	-0.729 (0.538)	-1.186** (0.541)	-1.625*** (0.626)
College degree	0.202 (0.678)	0.943 (0.882)	-1.407* (0.835)	-2.367** (0.969)
Employed	0.870 (0.725)	0.639 (0.939)	0.490 (0.895)	0.929 (1.037)
Parent	-0.028 (0.703)	-1.213 (0.915)	-0.800 (0.864)	-0.720 (0.999)
Constant	24.651*** (4.590)	28.087*** (5.696)	31.517*** (5.740)	40.559*** (6.647)
N	1,996	1,334	1,996	1,519
R ²	0.012	0.014	0.032	0.039

Notes: This table shows OLS regression estimates using respondents from wave 1. The dependent variable in each column is the absolute difference between the respondent's stated belief (behavior/norms) and the actual share. "Behavior belief" is the respondent's belief about the share of Americans who fight global warming. "Norms belief" is the respondent's belief about the share of Americans who think one should fight global warming. Columns 1 and 3 use the full sample, while columns 2 and 4 focus on the subset of respondents who underestimate the actual shares.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.6: Relationship of abstract and specific perceived norm measures

	(1) Restrict meat consumption	(2) Avoid taking flights	(3) Use car alternatives	(4) Use green electricity	(5) Adapt shopping behavior	(6) Political engagement
Panel A: Behavior						
Behavior belief	0.477*** (0.021)	0.362*** (0.022)	0.471*** (0.021)	0.421*** (0.021)	0.480*** (0.020)	0.468*** (0.020)
N	1,994	1,993	1,993	1,994	1,992	1,993
R ²	0.228	0.131	0.222	0.178	0.231	0.219
Panel B: Norms						
Norms belief	0.410*** (0.021)	0.340*** (0.021)	0.454*** (0.021)	0.416*** (0.020)	0.471*** (0.020)	0.448*** (0.020)
N	1,994	1,993	1,993	1,994	1,992	1,993
R ²	0.168	0.116	0.206	0.174	0.222	0.201

Notes: This table shows OLS regression estimates using respondents from wave 1. All coefficients can be interpreted as Pearson correlation coefficients. The dependent variables in Panel A are beliefs about the share of Americans who engage in the concrete climate-friendly behavior indicated in the column header. The dependent variables in Panel B are beliefs about the share of Americans who say that one should engage in the concrete climate-friendly behaviors. “Behavior belief” is the respondent’s belief about the share of Americans who fight global warming. “Norms belief” is the respondent’s belief about the share of Americans who think one should fight global warming. All beliefs are standardized to have a mean of zero and a standard deviation of one.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.7: Test of balance

	Means (std. dev.)			Differences (<i>p</i> -values)		
	Control group (C)	Behavior treatment (T ₁)	Norms treatment (T ₂)	T ₁ - C	T ₂ - C	T ₂ - T ₁
Behavior belief	52.096 (21.339)	51.627 (21.213)	51.644 (21.391)	-0.470 (0.486)	-0.452 (0.503)	-0.017 (0.980)
Norms belief	62.172 (21.357)	61.667 (21.535)	61.328 (21.948)	-0.505 (0.458)	-0.845 (0.217)	0.339 (0.621)
Altruism	-0.008 (0.982)	-0.024 (0.984)	0.032 (1.032)	-0.016 (0.600)	0.040 (0.206)	-0.057* (0.076)
Patience	-0.020 (0.993)	0.005 (0.989)	0.015 (1.019)	0.025 (0.424)	0.035 (0.265)	-0.010 (0.744)
Risk	-0.001 (0.989)	-0.005 (1.011)	0.006 (1.000)	-0.005 (0.887)	0.007 (0.827)	-0.011 (0.719)
Pos. reciprocity	-0.018 (1.024)	0.021 (0.983)	-0.002 (0.993)	0.039 (0.223)	0.016 (0.619)	0.023 (0.463)
Neg. reciprocity	-0.011 (0.999)	0.012 (0.978)	-0.001 (1.023)	0.023 (0.455)	0.011 (0.733)	0.013 (0.692)
Trust	-0.028 (1.001)	0.017 (1.000)	0.010 (0.999)	0.045 (0.156)	0.038 (0.229)	0.007 (0.825)
Rel. universalism	-0.027 (0.987)	0.021 (1.020)	0.006 (0.993)	0.047 (0.138)	0.032 (0.303)	0.015 (0.639)
Age	48.114 (17.727)	47.350 (17.055)	47.847 (17.438)	-0.763 (0.166)	-0.266 (0.632)	-0.497 (0.361)
Female	0.494 (0.500)	0.522 (0.500)	0.514 (0.500)	0.029* (0.071)	0.020 (0.202)	0.008 (0.593)
Log income	10.782 (0.882)	10.795 (0.879)	10.815 (0.858)	0.013 (0.645)	0.033 (0.236)	-0.020 (0.471)
College degree	0.473 (0.499)	0.479 (0.500)	0.457 (0.498)	0.007 (0.676)	-0.015 (0.335)	0.022 (0.166)
Employed	0.499 (0.500)	0.488 (0.500)	0.506 (0.500)	-0.012 (0.467)	0.007 (0.672)	-0.018 (0.248)
Democrat	0.528 (0.499)	0.535 (0.499)	0.539 (0.499)	0.007 (0.640)	0.011 (0.497)	-0.003 (0.833)
Northeast	0.170 (0.376)	0.165 (0.372)	0.174 (0.380)	-0.005 (0.692)	0.004 (0.717)	-0.009 (0.447)
Midwest	0.204 (0.403)	0.211 (0.408)	0.216 (0.411)	0.007 (0.602)	0.012 (0.362)	-0.005 (0.697)
South	0.390 (0.488)	0.385 (0.487)	0.365 (0.482)	-0.005 (0.743)	-0.025 (0.105)	0.020 (0.196)
Parent	0.562 (0.496)	0.557 (0.497)	0.550 (0.498)	-0.005 (0.762)	-0.012 (0.441)	0.007 (0.640)
<i>p</i> -value of joint <i>F</i> -test				0.426	0.684	0.425
Observations	1,987	1,995	2,018	3,982	4,005	4,013

Notes: Columns 1–3 show the means and standard deviations of respondent covariates in the different treatments of wave 2. Columns 4–6 show differences in means between the groups indicated in the column header together with *p*-values in parentheses. The *p*-values of the joint *F*-test are determined by regressing the treatment indicator on the vector of demographic controls. The *F*-test tests the joint hypothesis that none of the covariates predicts treatment assignment.

Covariates “Behavior belief” and “Norms belief” are the perceived social norm measures, ranging from 0 to 100. Economic preferences (altruism, patience, risk, pos. reciprocity, neg. reciprocity, trust) and moral universalism (rel. universalism) are standardized. “Female”, “Employed”, “Democrat”, “Parent”, and the three census region dummies are binary indicators.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.8: Treatment effects on climate donations and posterior beliefs: No controls

	(1) Donation (\$)	(2) Behavior belief (post.)	(3) Norms belief (post.)
Behavior treatment	12.852*** (4.824)	0.285*** (0.031)	0.244*** (0.031)
Norms treatment	17.485*** (4.857)	0.374*** (0.031)	0.355*** (0.031)
N	5,991	5,988	5,976
Control group mean	249.31	0	0
z-scored	No	Yes	Yes

Notes: This table shows OLS regression estimates using respondents from wave 2. “Behavior treatment” is a binary indicator taking value one for respondents who received information about the share of Americans who try to fight global warming. “Norms treatment” is a binary indicator taking value one for respondents who received information about the share of Americans who say that one should try to fight global warming. “Behavior belief” is an index of six post-treatment beliefs about the share of Americans engaging in concrete climate-friendly behaviors to fight global warming. “Norms belief” is an index of six post-treatment beliefs about the share of Americans who say that one should engage in concrete climate-friendly behaviors to fight global warming. Both indices are standardized to have a mean of zero and a standard deviation of one in the control group.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.9: Treatment effect heterogeneity for post-treatment beliefs: Prior above/below actual share

	Prior below actual share		Prior above actual share	
	(1) Behavior belief (post.)	(2) Norms belief (post.)	(3) Behavior belief (post.)	(4) Norms belief (post.)
Panel A: Behavior treatment				
Behavior treatment	0.401*** (0.033)	0.366*** (0.033)	0.062 (0.056)	0.003 (0.054)
N	2,577	2,573	1,397	1,398
Control group mean	-0.23	-0.22	0.42	0.41
Controls	Yes	Yes	Yes	Yes
Panel B: Norms treatment				
Norms treatment	0.422*** (0.032)	0.433*** (0.032)	0.195*** (0.072)	0.078 (0.068)
N	3,054	3,043	945	946
Control group mean	-0.13	-0.15	0.42	0.49
Controls	Yes	Yes	Yes	Yes

Notes: This table shows OLS regression estimates using respondents from wave 2. The dependent variables are post-treatment beliefs. It is regressed on binary indicators that take the value of 1 for respondents in the behavior treatment and norms treatment, respectively. We run separate analyses for respondents with prior norm perceptions strictly below the actual share (columns 1-2) and equal to or above the actual share (columns 3-4). All regressions include the set of controls described in Table 2.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.10: Treatment effect heterogeneity for post-treatment beliefs: Median-split by prior beliefs among respondents with prior belief below the actual share

	(1) Behavior belief (post.)	(2) Norms belief (post.)
Behavior treatment	0.183*** (0.048)	
Low behavior prior	-0.572*** (0.045)	
Behavior treatment x Low behavior prior	0.424*** (0.065)	
Norms treatment		0.338*** (0.045)
Low norms prior		-0.559*** (0.042)
Norms treatment x Low norms prior		0.210*** (0.062)
N	2,577	3,043
Controls	Yes	Yes

Notes: This table shows OLS regression estimates using respondents from wave 2. The dependent variables are post-treatment beliefs. We run separate regressions for respondents with prior behavior perceptions strictly below the actual share (columns 1) and respondents with prior norm perceptions strictly below the actual share (column 2). Regression include a binary indicator that take the value of 1 for respondents in the behavior treatment and norms treatment, respectively. This indicator is interacted with a binary indicator for whether the respondent has a below-median prior belief (among all respondents who underestimate the actual shares). All regressions include the set of controls described in Table 2.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.11: Treatment effect heterogeneity by political affiliation

	Dependent variable:		
	(1) Donation (\$)	(2) Policy support	(3) Support activism
Panel A: Behavior treatment			
Treatment	18.114** (7.363)	0.155*** (0.048)	0.072 (0.045)
Treatment x Democrat	-10.940 (9.495)	-0.119** (0.054)	-0.051 (0.056)
Democrat	75.232*** (6.839)	1.029*** (0.040)	0.911*** (0.041)
N	3,978	3,981	3,977
Controls	No	No	No
Panel B: Norms treatment			
Treatment	16.904** (7.507)	0.114** (0.049)	0.045 (0.044)
Treatment x Democrat	-0.604 (9.557)	-0.081 (0.055)	-0.058 (0.057)
Democrat	75.232*** (6.839)	1.029*** (0.040)	0.911*** (0.041)
N	4,000	4,005	4,002
Controls	No	No	No

Notes: This table shows OLS regressions from wave 2. The dependent variables are indicated by the column header. It is regressed on a treatment dummy for the behavior treatment (Panel A) and the norm treatment (Panel B), respectively, a binary indicator for whether respondents identify as Democrats, and the interaction of this indicator with the respective treatment indicator.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.12: Treatment effect heterogeneity by political affiliation: Robustness to prior x treatment interaction

	Dependent variable:		
	(1) Donation (\$)	(2) Policy support	(3) Support activism
Panel A: Behavior treatment			
Treatment	28.457** (13.787)	0.159* (0.082)	-0.056 (0.076)
Treatment x Democrat	-9.206 (9.456)	-0.103* (0.054)	-0.043 (0.055)
Democrat	73.085*** (6.845)	0.996*** (0.040)	0.859*** (0.040)
N	3,978	3,981	3,977
Controls	Yes	Yes	Yes
Treatment x Prior	Yes	Yes	Yes
Panel B: Norms treatment			
Treatment	34.966** (15.344)	0.200** (0.094)	0.075 (0.084)
Treatment x Democrat	1.153 (9.520)	-0.074 (0.054)	-0.059 (0.055)
Democrat	70.477*** (6.820)	0.985*** (0.039)	0.843*** (0.040)
N	4,000	4,005	4,002
Controls	Yes	Yes	Yes
Treatment x Prior	Yes	Yes	Yes

Notes: This table shows OLS regressions from wave 2. The dependent variables are indicated by the column header. It is regressed on a treatment dummy for the behavior treatment (Panel A) and the norm treatment (Panel B), respectively, a binary indicator for whether respondents identify as Democrats, and the interaction of this indicator with the respective treatment indicator. All regressions include sociodemographic controls and the interaction between the prior belief and the respective treatment.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.13: Baseline differences in donations: “Climate change skeptics”

	Dependent variable: Donation (\$)				
	(1)	(2)	(3)	(4)	(5)
Panel A: No controls					
No trust in science	-107.464*** (6.736)				
No scientific consensus		-102.040*** (7.209)			
Not concerned			-150.412*** (7.239)		
Not a threat				-140.521*** (7.223)	
Caused by nature					-142.019*** (7.365)
Constant	296.258*** (4.064)	284.234*** (3.944)	292.462*** (3.603)	290.968*** (3.681)	289.982*** (3.637)
N	1,987	1,987	1,987	1,987	1,987
Panel B: Controls					
No trust in science	-56.692*** (7.491)				
No scientific consensus		-64.028*** (7.422)			
Not concerned			-105.933*** (8.319)		
Not a threat				-96.897*** (8.027)	
Caused by nature					-101.338*** (8.174)
Constant	243.600*** (47.752)	236.416*** (47.364)	256.064*** (46.300)	219.876*** (46.711)	260.010*** (46.361)
N	1,977	1,977	1,977	1,977	1,977

Notes: This table shows OLS regression estimates where the dependent variable is the donation decision (in \$) using control group respondents from wave 2. Panel A presents estimates without the inclusion of controls for economic preferences, moral values, perceived social norms, and sociodemographic controls. Panel B includes these controls. “No trust in science” means that the respondent trusts climate scientists “a moderate amount”, “a little” or not at all (on a five-point Likert scale). “No scientific consensus” means that the respondent thinks that most scientists think that global warming is not happening or that there is no consensus among scientists. “Not concerned” means that the respondent is “not very worried” or “not at all worried” about global warming (on a four-point Likert scale). “Not a threat” means that the respondent thinks that global warming will do “only a little” or no harm at all (on a four-point Likert scale). “Caused by nature” means that the respondent thinks that global warming is caused by natural activities.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.14: Treatment effect heterogeneity: Climate change “skeptics”

	Dependent variable: Donation (\$)				
	Interactant:				
	(1) No trust in science	(2) No scientific consensus	(3) Not concerned	(4) Not a threat	(5) Caused by nature
Panel A: Behavior treatment					
Treatment (a)	2.733 (5.661)	1.335 (5.392)	1.004 (5.007)	1.895 (5.085)	0.122 (5.082)
Treatment x Interactant (b)	18.268* (9.357)	22.561** (10.126)	33.200*** (10.410)	29.943*** (10.330)	38.333*** (10.466)
Interactant	-91.364*** (7.145)	-82.718*** (7.472)	-140.489*** (7.751)	-128.326*** (7.710)	-127.592*** (7.865)
Linear combination (a + b)	21.001*** (7.444)	23.896*** (8.568)	34.204*** (9.121)	31.837*** (8.981)	38.455*** (9.144)
N	3,978	3,978	3,978	3,978	3,978
Controls	Yes	Yes	Yes	Yes	Yes
Panel B: Norms treatment					
Treatment (a)	13.000** (5.667)	8.245 (5.460)	10.241** (4.987)	9.397* (5.069)	11.639** (5.053)
Treatment x Interactant (b)	7.751 (9.353)	21.274** (10.044)	14.928 (10.406)	14.560 (10.398)	14.569 (10.386)
Interactant	-89.976*** (7.140)	-80.385*** (7.465)	-139.925*** (7.742)	-127.516*** (7.726)	-128.427*** (7.852)
Linear combination (a + b)	20.751*** (7.442)	29.519*** (8.431)	25.169*** (9.136)	23.957*** (9.084)	26.208*** (9.082)
N	4,000	4,000	4,000	4,000	4,000
Controls	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS regressions from wave 2. The dependent variable is the donation to the climate charity (\$). It is regressed on a treatment dummy for the behavior treatment (Panel A) and the norm treatment (Panel B), respectively, an interactant that varies across columns, and its interaction with the treatment dummy. Interactants are indicated by the column header. Each interactant is a binary variable taking value one. “No trust in science” means that the respondent trusts climate scientists “a moderate amount”, “a little” or not at all (on a five-point Likert scale). “No scientific consensus” means that the respondent thinks that most scientists think that global warming is not happening or that there is no consensus among scientists. “Not concerned” means that the respondent is “not very worried” or “not at all worried” about global warming (on a four-point Likert scale). “Not a threat” means that the respondent thinks that global warming will do “only a little” or no harm at all (on a four-point Likert scale). “Caused by nature” means that the respondent thinks that global warming is caused by natural activities.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.15: Treatment effect heterogeneity: Climate change “skeptics” – Robustness to controlling for the interaction between treatment and prior beliefs

	Dependent variable: Donation (\$)				
	Interactant:				
	(1) No trust in science	(2) No scientific consensus	(3) Not concerned	(4) Not a threat	(5) Caused by nature
Panel A: Behavior treatment					
Treatment (a)	9.683 (13.391)	9.228 (13.147)	12.670 (12.671)	13.362 (12.741)	13.353 (12.617)
Treatment x Interactant (b)	17.090* (9.420)	21.511** (10.142)	32.559*** (10.464)	29.391*** (10.379)	38.440*** (10.424)
Interactant	-89.111*** (7.202)	-80.718*** (7.502)	-138.804*** (7.782)	-126.678*** (7.742)	-127.584*** (7.823)
Linear combination (a + b)	26.772** (13.242)	30.739** (13.961)	45.229*** (13.740)	42.753*** (13.726)	51.793*** (13.967)
N	3,978	3,978	3,978	3,978	3,978
Controls	Yes	Yes	Yes	Yes	Yes
Treatment x Prior	Yes	Yes	Yes	Yes	Yes
Panel B: Norms treatment					
Treatment (a)	27.580* (15.650)	18.851 (15.657)	22.250 (15.042)	25.774* (15.231)	26.725* (14.881)
Treatment x Interactant (b)	5.596 (9.481)	18.748* (10.173)	13.119 (10.688)	13.138 (10.649)	13.001 (10.508)
Interactant	-84.081*** (7.214)	-74.126*** (7.569)	-134.167*** (7.935)	-121.945*** (7.871)	-123.874*** (7.927)
Linear combination (a + b)	33.176** (14.660)	37.599** (15.099)	35.370** (14.536)	38.912*** (14.745)	39.726*** (14.640)
N	4,000	4,000	4,000	4,000	4,000
Controls	Yes	Yes	Yes	Yes	Yes
Treatment x Prior	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS regressions from wave 2. The dependent variable is the donation to the climate charity (\$). It is regressed on a treatment dummy for the behavior treatment (Panel A) and the norm treatment (Panel B), respectively, an interactant that varies across columns, and its interaction with the treatment dummy. Interactants are indicated by the column header. Each interactant is a binary variable taking value one. “No trust in science” means that the respondent trusts climate scientists “a moderate amount”, “a little” or not at all (on a five-point Likert scale). “No scientific consensus” means that the respondent thinks that most scientists think that global warming is not happening or that there is no consensus among scientists. “Not concerned” means that the respondent is “not very worried” or “not at all worried” about global warming (on a four-point Likert scale). “Not a threat” means that the respondent thinks that global warming will do “only a little” or no harm at all (on a four-point Likert scale). “Caused by nature” means that the respondent thinks that global warming is caused by natural activities. All regressions include the corresponding prior belief and the interaction between the treatment indicator and the prior belief.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.16: Moral universalism and climate change skepticism

	Dependent variable: Relative moral universalism (z-scored)				
	(1)	(2)	(3)	(4)	(5)
Low trust in science	-0.718*** (0.040)				
No scientific consensus		-0.710*** (0.040)			
Not concerned			-0.747*** (0.042)		
No harm				-0.790*** (0.040)	
Nature-caused					-0.775*** (0.041)
N	1,980	1,980	1,980	1,980	1,980

Notes: This table shows OLS regression estimates using control group respondents from wave 2. It regresses the standardized moral universalism index on various binary indicators of climate change skepticism. “Low trust in science” means that the respondent trusts climate scientists “a moderate amount”, “a little” or not at all (on a five-point Likert scale). “No scientific consensus” means that the respondent thinks that most scientists think that global warming is not happening or that there is no consensus among scientists. “Not concerned” means that the respondent is “not very worried” or “not at all worried” about global warming (on a four-point Likert scale). “Not a threat” means that the respondent thinks that global warming will do “only a little” or no harm at all (on a four-point Likert scale). “Caused by nature” means that the respondent thinks that global warming is caused by natural activities.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Table A.17: Treatment effect heterogeneity for moral universalism

	Dependent variable: Donations (\$)					
	(1)	(2)	(3)	(4)	(5)	(6)
Behavior treatment	10.622** (4.620)	20.974* (12.679)	11.185 (12.633)			
Norms treatment				15.393*** (4.636)	34.641** (14.451)	27.885* (14.734)
Behavior treatment x Moral universalism	-9.181** (4.576)	-8.964** (4.562)	-1.739 (4.726)			
Norms treatment x Moral universalism				-5.718 (4.629)	-5.325 (4.615)	-3.979 (4.785)
Behavior treatment x Behavior belief		-19.474 (22.439)	-21.660 (21.556)			
Norms treatment x Norms belief					-29.949 (21.984)	-23.531 (21.313)
Behavior treatment x Skeptic			37.601*** (11.151)			
Norms treatment x Skeptic						11.054 (11.251)
Moral universalism	33.497*** (3.510)	33.575*** (3.497)	21.540*** (3.529)	32.878*** (3.523)	32.322*** (3.497)	21.415*** (3.523)
N	3,963	3,963	3,963	3,987	3,987	3,987
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS regression estimates using respondents from wave 2. Columns 1–3 use respondents assigned to the control group or the behavior treatment, while columns 4–6 use respondents assigned to the control group and the norms treatment. The dependent variables are donations (in \$). It is regressed on binary indicators that take the value of 1 for respondents in the behavior treatment. “Skeptic” is a binary indicator taking value 1 for respondents who do not believe in human-caused climate change. All regressions include the set of controls described in Table 2. The regressions also include the behavior belief (columns 2 and 3) and the norms belief (columns 5 and 6) as well as the a dummy for climate change skepticism (columns 3 and 6).

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

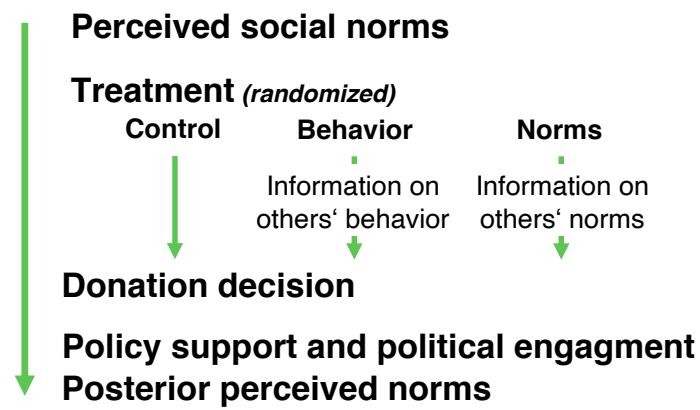
Table A.18: Treatment effect heterogeneity for political outcomes: Prior above/below actual share

	Prior below actual share			Prior above actual share		
	(1) Policies	(2) Activism	(3) Both	(4) Policies	(5) Activism	(6) Both
Panel A: Behavior treatment						
Behavior treatment	0.097*** (0.033)	0.001 (0.033)	0.039 (0.032)	0.063 (0.043)	0.106** (0.047)	0.096** (0.045)
N	2,582	2,577	2,577	1,399	1,400	1,399
Control group mean	-0.03	-0.06	-0.05	0.06	0.10	0.09
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Panel B: Norms treatment						
Norms treatment	0.078** (0.030)	0.005 (0.030)	0.035 (0.029)	0.036 (0.052)	0.033 (0.059)	0.036 (0.055)
N	3,058	3,056	3,056	947	946	946
Control group mean	-0.06	-0.05	-0.06	0.18	0.17	0.19
Controls	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table shows OLS regression estimates using respondents from wave 2. The dependent variables are support for climate policies (columns 1, 4), willingness to engage in political activism (column 2, 5), and the index using both measures (columns 3, 6). It is regressed on binary indicators that take the value of 1 for respondents in the behavior treatment and norms treatment, respectively. We run separate analyses for respondents with prior norm perceptions strictly below the actual share (columns 1-3) and equal to or above the actual share (columns 4-6). All regressions include the set of controls described in Table 2.

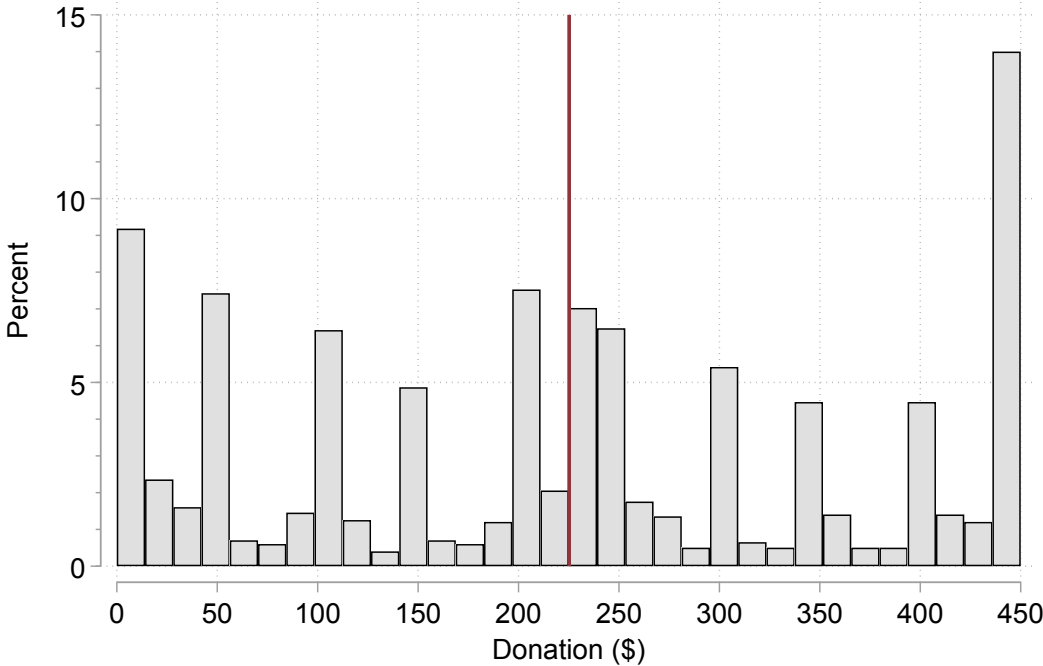
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Robust standard errors in parentheses.

Figure A.1: Structure of experiment



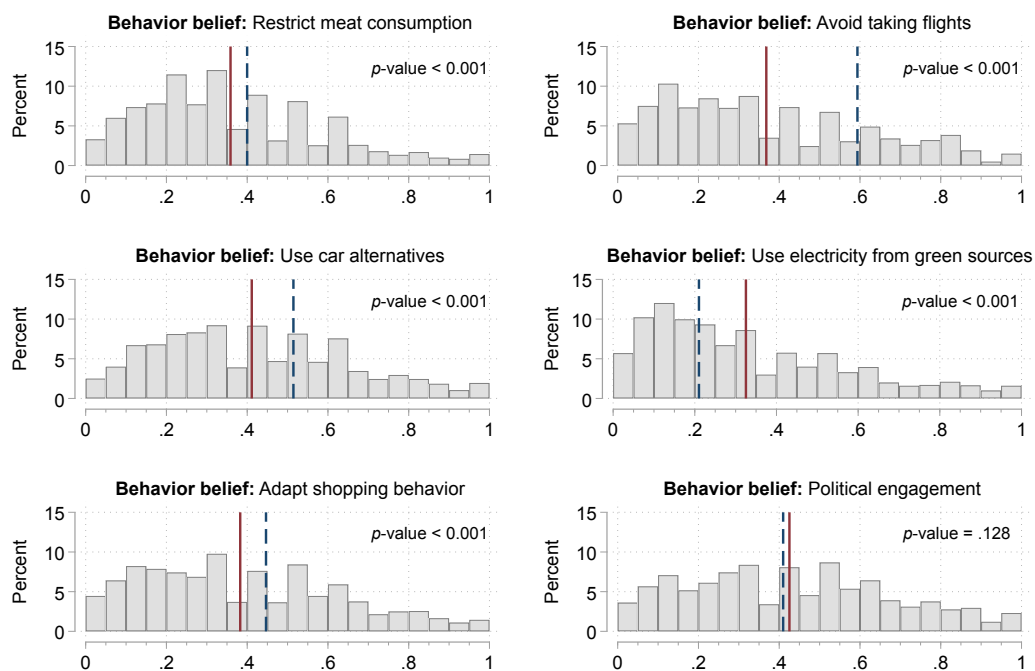
Notes: This figure provides an overview of the structure of the experiment.

Figure A.2: The distribution of individual willingness to fight global warming



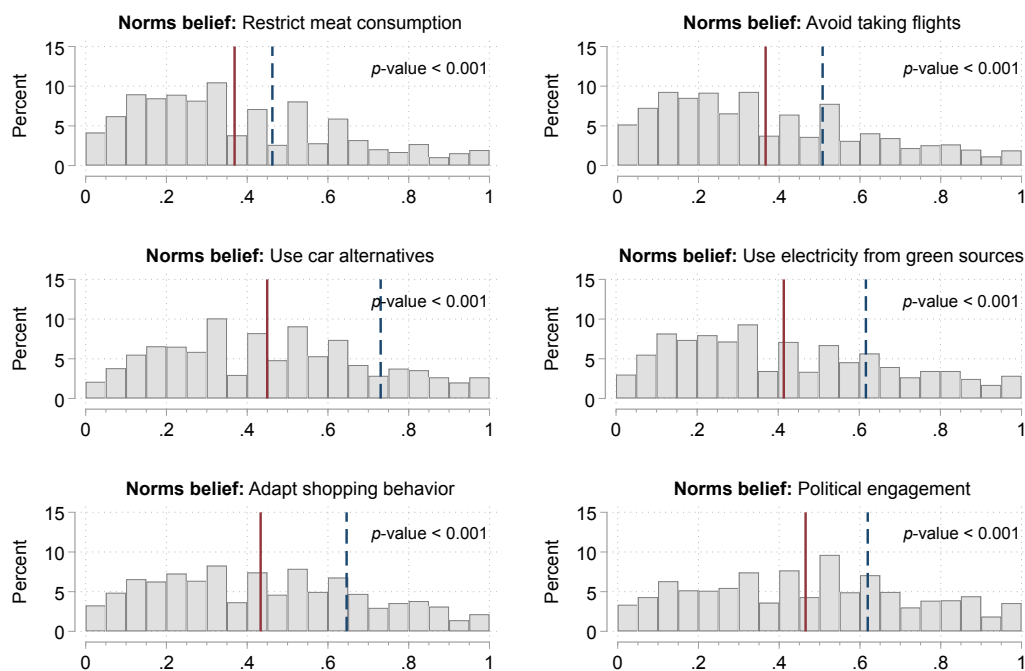
Notes: This figure shows the distribution of the monetary amounts donated to the climate charity in wave 1. The average donation is indicated by the vertical red line.

Figure A.3: Perceived prevalence of concrete climate-friendly behaviors



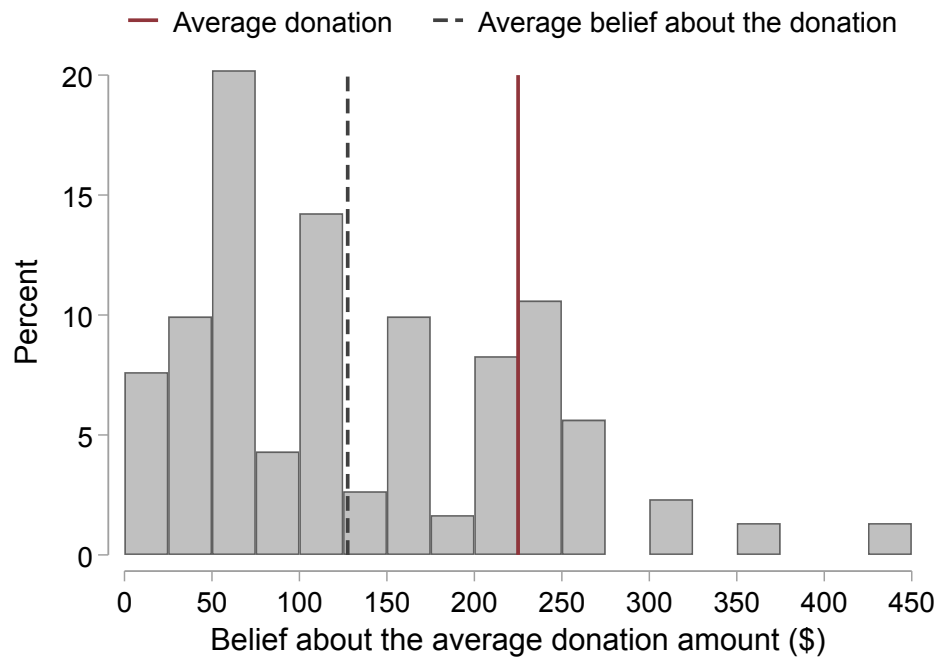
Notes: This figure shows the distribution of behavior beliefs in wave 1 for concrete climate-friendly behaviors. Each panel shows the distribution of people's beliefs about the share of Americans who say that they engage in the specific climate-friendly behavior indicated in the title of the panel. The solid red line indicates the average belief. The dashed blue line indicates the actual share of Americans engaging the behavior.

Figure A.4: Perceived prevalence of norms for concrete climate-friendly behavior



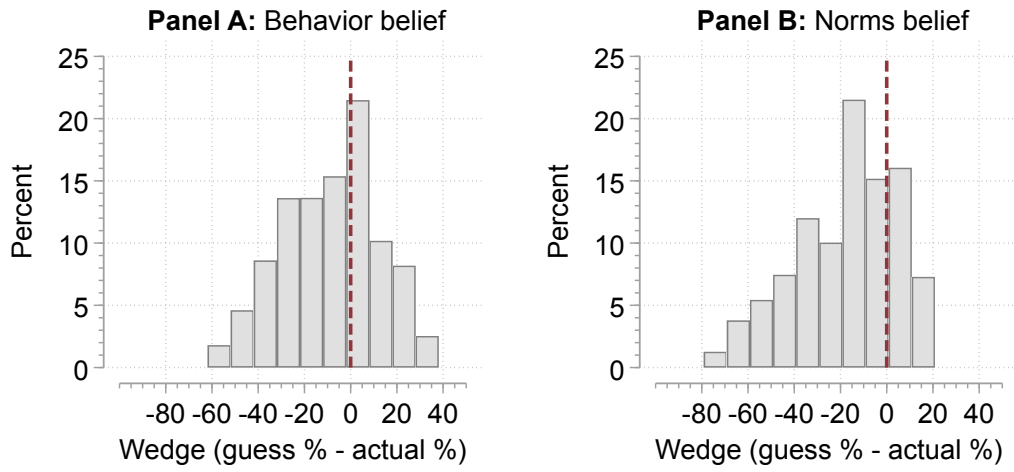
Notes: This figure shows the distribution of norms beliefs in wave 1 for concrete climate-friendly behaviors. Each panel shows the distribution of people's beliefs about the share of Americans who say that one *should* engage in the specific climate-friendly behavior. The solid red line indicates the average belief. The dashed blue line indicates the actual share of Americans saying that one should engage in the behavior indicated in the title of the panel.

Figure A.5: Auxiliary experiment: Beliefs about the donation behavior of others



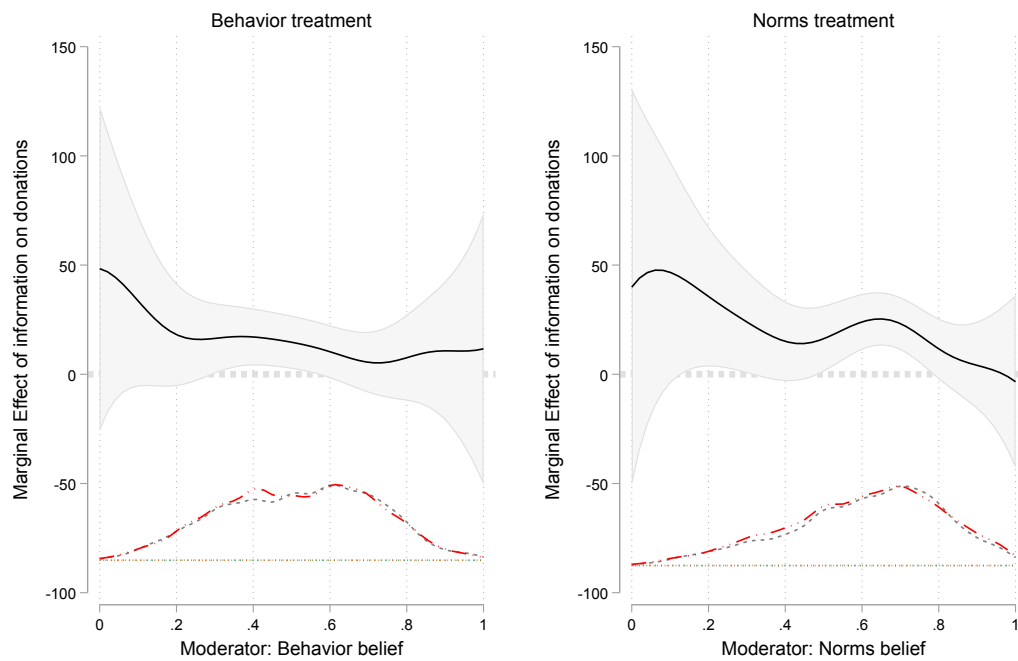
Notes: This figure uses data from an auxiliary experiment conducted on *Prolific* in November 2022 ($N = 302$). We elicit respondents' beliefs about the average donation amount in the main experiment. The figure plots the distribution of beliefs about the average donation amount. The average belief is \$127.6 compared to the actual average donation of \$225 in the wave 1 of the main experiment. 86% of respondents underestimate the actual donation amount.

Figure A.6: Wedge in beliefs about social norms



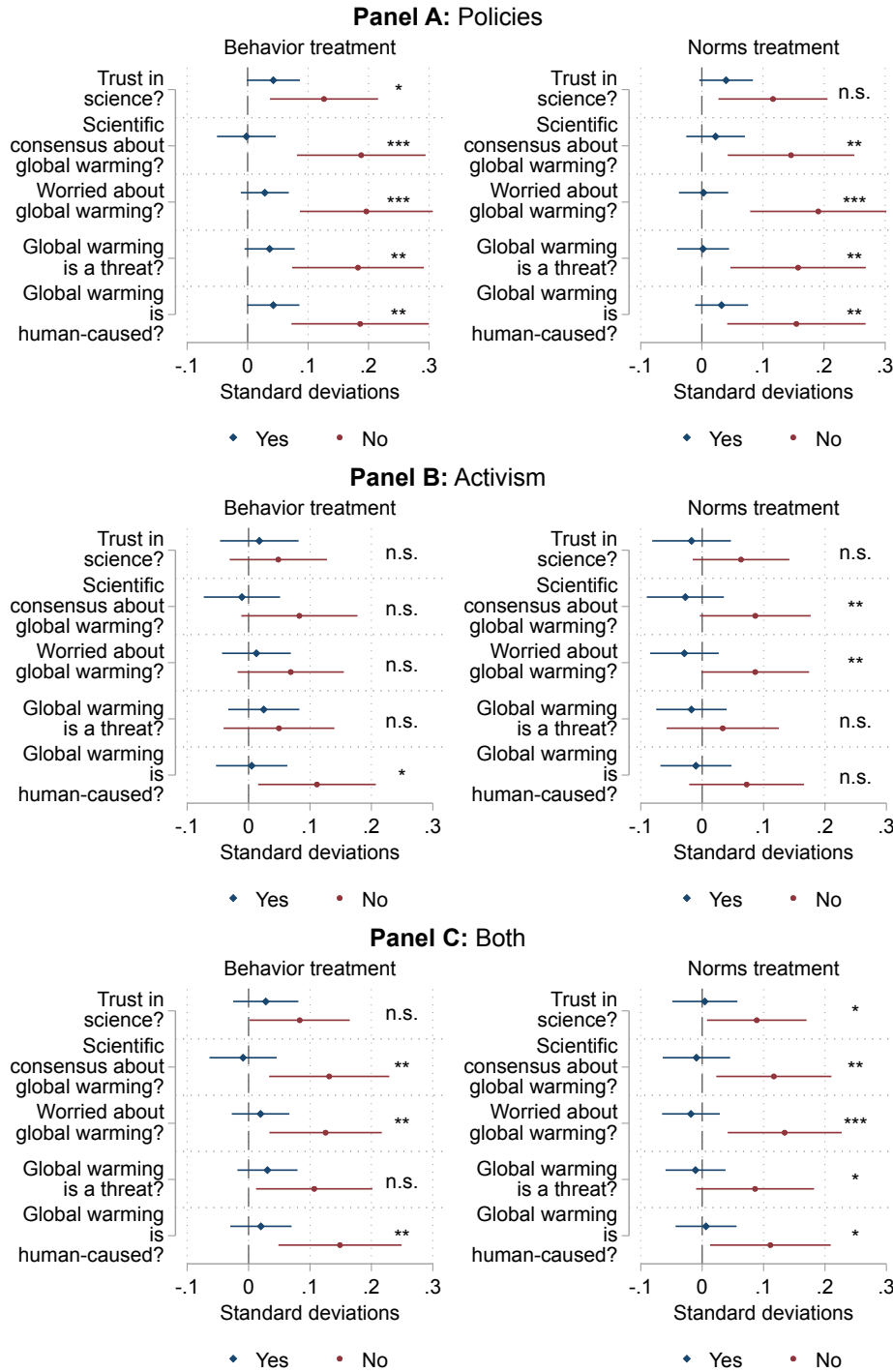
Notes: Using respondents from wave 2, this figure shows the distribution of the wedge between the respondent's perceived social norms and the actual shares in wave 1. Panel A shows people's belief about the share of Americans who say that they try to fight global warming. Panel B shows people's belief about the share of Americans who say that one should fight global warming. The red vertical line indicates the actual shares from wave 1.

Figure A.7: Treatment effect heterogeneity by perceived social norms: Non-parametric estimates



Notes: This figure shows the results from a non-linear interaction analysis using the *interflex* package (Xu et al., 2017; Hainmueller et al., 2019) and restricting the sample to respondents from wave 2. The left panel excludes respondents in the norms treatment, while the right panel excludes respondents in the behavior treatment. The dashed lines at the bottom of each panel plot the distribution of the pre-treatment belief. 95% confidence intervals using robust standard errors are shown. Both panels show results without including additional controls.

Figure A.8: Heterogeneity by “climate change skeptics”: Political outcomes



Notes: This figure shows treatment effects in different subsamples using respondents from wave 2. Panel A shows treatment effects on the policy support index, Panel B shows treatment effects of the action index, and Panel C shows treatment effects on the joint index. 95% confidence intervals are shown. Each panel shows estimates for the subsample of climate change skeptics – e.g., those who have no trust in science or do not believe in human-caused global warming – and the subsample of respondents who are not skeptical of climate change. See Figure 3 for an explanation of the labels. For each sample split, we indicate the level of significance of a test of equality of coefficients.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$, n.s. $p \geq 0.10$.

Appendix B Questionnaire

This appendix presents the main survey blocks, following the order of exposition in the paper. The full questionnaire containing all questions administered as part of this study can be downloaded from <https://osf.io/chvy6/>.

B.1 Attention screener

The next question is about the following problem. In questionnaires like ours, sometimes there are participants who do not carefully read the questions and just quickly click through the survey. This compromises the results of research studies. **To show that you are reading the survey carefully, please choose both “Very strongly interested” and “Not at all interested” as your answer to the next question.**

Given the above, how interested are you in politics?

- a) Very strongly interested
- b) Very interested
- c) A little bit interested
- d) Not very interested
- e) Not at all interested

Only participants who select both (a) and (e) pass this attention screener.

B.2 Measuring individual willingness to fight climate change

A decision about money

Please pay special attention to the next question in which you will make a decision about money. We will randomly select 25 respondents. If you are among them, your decision will be a real decision. The decision will be implemented and you can receive up to \$450.

Your decision

Here is the decision: You can divide \$450 between yourself and a charitable organization that fights global warming. The amount that you keep for yourself will be added to your account. The amount that you donate will go to the award-winning charity *atmosfair*. *atmosfair* actively contributes to CO₂ mitigation by promoting, developing and financing

renewable energies worldwide. In this way, a donation saves CO_2 that would otherwise be created by fossil fuels. *atmosfair* spends around \$12 million per year to fight global warming and uses less than 5% of donated funds to cover administrative costs. You can find more information on *atmosfair* [here](#).

It costs about \$450 to offset the yearly CO_2 emissions of a typical US citizen. This number is calculated as follows: It costs about \$28 to prevent 1 ton of CO_2 emissions. The World Bank estimates that a typical US citizen causes about 16 tons of CO_2 emissions per year.

How much of the \$450 would you like to donate to *atmosfair*?

B.3 Introducing bonus scheme

Bonus payment possible

There are several questions in this survey, in which we will ask you to guess how other respondents answered a question. These questions are flagged with the sign:



You can earn a bonus of \$1. This works as follows: We will randomly select one of the flagged questions. Your response to this question is considered as correct if it differs at most by three from the correct number you are asked to guess. If your response to this question is correct, \$1 will be added to your account.

B.4 Measuring perceived social norms

Do you try to fight global warming?

[Yes/No]

Do you think that people in the United States should try to fight global warming?

[Yes/No]

[PAGE BREAK]

The questions on this page are bonus questions. This means that you can earn additional money if you answer them correctly.



As part of this research project, we recently surveyed many people in the United States and asked them the same questions. Respondents come from all parts of the population and their responses represent the views and attitudes of people in the United States. What do you think? Out of 100 people we asked, how many stated that...

- a) ... they try to fight global warming?
- b) ... they think that people in the United States should try to fight global warming?

B.5 Treatments: Shifting perceived social norms

B.5.1 Behavior treatment

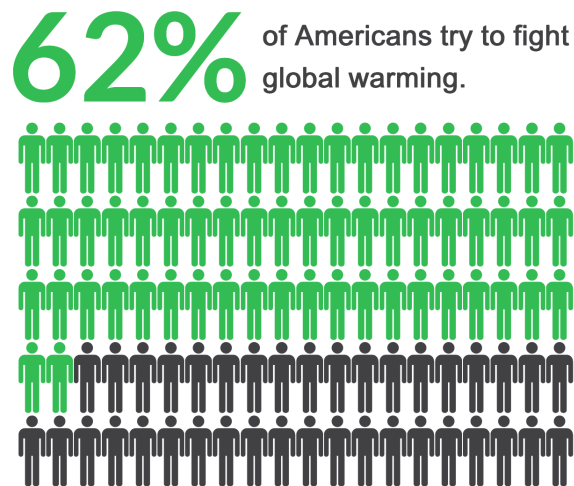
What do other people in the United States do?

We recently surveyed 2,000 people in the United States and asked them whether they try to fight global warming. Respondents come from all parts of the population and their responses represent the views and attitudes of people in the United States. On the next page, you will learn how they responded. Please read the information carefully.

[PAGE BREAK]

We asked 2,000 Americans: Do you try to fight global warming? Yes or no?

Here are the results:



B.5.2 Norms treatment

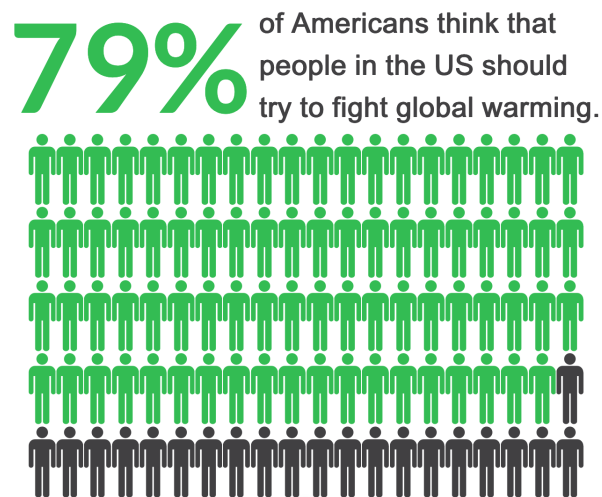
What do other people in the United States think?

We recently surveyed 2,000 people in the United States and asked them whether they think people in the US should try to fight global warming. Respondents come from all parts of the population and their responses represent the views and attitudes of people in the United States. On the next page, you will learn how they responded. Please read the information carefully.

[PAGE BREAK]

We asked 2,000 Americans: Do you think that people in the United States should try to fight global warming? Yes or no?

Here are the results:



B.6 Measuring posterior beliefs

The questions on this page are bonus questions. This means that you can earn additional money if you answer them correctly.



As part of this research project, we recently surveyed many people in the United States and asked them the same questions. Respondents come from all parts of the population and their responses represent the views and attitudes of people in the United States.

We asked respondents to state whether they have taken different actions to fight global warming over the last year.

What do you think? Out of 100 people we asked, how many stated that...

- a) ... restrict their meat consumption?
- b) ... avoid taking flights?
- c) ... regularly use environmentally-friendly alternatives to their private car such as walking, cycling, taking public transport or car-sharing?
- d) ... receive electricity only from green/renewable sources (e.g., solar energy or wind power)?
- e) ... adapt their shopping behavior to the carbon footprint of products?
- f) ... politically support the fight against global warming, e.g. participate in a demonstration, sign a letter, or support a political organization?

[PAGE BREAK]

Do you think that people in the United states **should**...

- a) ... restrict their meat consumption?
- b) ... avoid taking flights?
- c) ... regularly use environmentally-friendly alternatives to their private car such as walking, cycling, taking public transport or car-sharing?

- d) ... receive electricity only from green/renewable sources (e.g., solar energy or wind power)?
- e) ... adapt their shopping behavior to the carbon footprint of products?
- f) ... politically support the fight against global warming, e.g. participate in a demonstration, sign a letter, or support a political organization?

[PAGE BREAK]

The questions on this page are bonus questions. This means that you can earn additional money if you answer them correctly.



What do you think? Out of 100 people we asked the same questions, how many stated that they think that people in the United States should...

- a) ... restrict their meat consumption?
- b) ... avoid taking flights?
- c) ... regularly use environmentally-friendly alternatives to their private car such as walking, cycling, taking public transport or car-sharing?
- d) ... receive electricity only from green/renewable sources (e.g., solar energy or wind power)?
- e) ... adapt their shopping behavior to the carbon footprint of products?
- f) ... politically support the fight against global warming, e.g. participate in a demonstration, sign a letter, or support a political organization?

B.7 Measuring climate change skepticism

In general, how much do you trust scientists who do research on global warming?

- a) A great deal
- b) A lot
- c) A moderate amount
- d) A little
- e) Not at all

Which comes closest to your own view?

- a) Most scientists think global warming is happening.
- b) There is a lot of disagreement among scientists about whether or not global warming is happening.
- c) Most scientists think global warming is not happening.

How worried are you about global warming?

- a) Very worried
- b) Somewhat worried
- c) Not very worried
- d) Not at all worried

How much do you think global warming will harm people in the United States?

- a) Not at all
- b) Only a little
- c) A moderate amount
- d) A great deal

Do you think that global warming is mainly...?

- a) a result of human activities
- b) a result of natural causes

B.8 Measuring policy support and political activism

Taken from the detailed politics module developed as part of the Climate Change in the American Mind Project (Howe et al., 2015).

Policy support

How much do you support or oppose the following policies?

Strongly support / Somewhat support / Somewhat oppose / Strongly oppose

- a) Fund more research into renewable energy sources, such as solar and wind power.
- b) Regulate carbon dioxide (the primary greenhouse gas) as a pollutant.
- c) Set strict carbon dioxide emission limits on existing coal-fired power plants to reduce global warming and improve public health. Power plants would have to reduce their emissions and/or invest in renewable energy and energy efficiency. The cost of electricity to consumers and companies would likely increase.
- d) Require fossil fuel companies to pay a carbon tax and use the money to reduce other taxes (such as income tax) by an equal amount.
- e) Require electric utilities to produce at least 20% of their electricity from wind, solar, or other renewable energy sources, even if it costs the average household an extra \$100 a year.
- f) Provide tax rebates for people who purchase energy-efficient vehicles or solar panels.

How much do you agree or disagree with the following statements?

Strongly agree / Somewhat agree / Somewhat disagree / Strongly disagree

- a) Schools should teach our children about the causes, consequences, and potential solutions to global warming.

Political activism

How likely would you be to do each of the following things?

Definitely would / Probably would / Probably would not / Definitely would not

- a) Vote for a candidate for public office because of their position on global warming.
- b) Publicly display t-shirt, bumper sticker, button, wrist band, or sign about global warming.
- c) Donate money to an organization working on global warming.
- d) Volunteer your time to an organization working on global warming.
- e) Write letters, email, or phone government officials about global warming.
- f) Meet with an elected official or their staff about global warming.
- g) Support an organization engaging in non-violent civil disobedience against corporate or government activities that make global warming worse.
- h) Personally engage in non-violent civil disobedience (e.g., sit-ins, blockades, or trespassing) against corporate or government activities that make global warming worse.
- i) Attend a political rally, speech, or organized protest about global warming.
- j) Write a letter to the editor of a newspaper or magazine or call a live radio or TV show to express an opinion about global warming.
- k) Share information about global warming on social media.

Appendix C Construction of Variables

C.1 Measuring economic preferences

We administer the Global Preferences Survey (GPS) and follow the methodology described in Falk et al. (2018) to obtain detailed individual-level measures of economic preferences. More information on the construction of the variables can be found below.

1. *Patience.* The measure of patience (or time preference) is derived from the combination of responses to two survey measures, one with a quantitative and one with a qualitative format. The quantitative survey measure consists of a series of five interdependent hypothetical binary choices between immediate and delayed financial rewards. In each of the five questions, participants have to decide between receiving a payment today or a larger payment in 12 months. The qualitative measure of patience is given by the respondents' self-assessment regarding their willingness to wait on an eleven-point Likert scale, asking "how willing are you to give up something that is beneficial for you today in order to benefit more from that in the future?".
2. *Risk Taking.* Risk preferences are also elicited through a series of related quantitative questions as well as one qualitative question. Just as with patience, the quantitative measure consists of a series of five binary choices. Choices are between a fixed lottery, in which the individual could win x or zero, and varying sure payments, y . The qualitative item asks for the respondents' self-assessment of their willingness to take risks on an eleven-point scale ("In general, how willing are you to take risks?").
3. *Positive Reciprocity.* Positive reciprocity is measured using one quantitative item and one qualitative question. First, respondents are presented a choice scenario in which they are asked to imagine that they got lost in an unfamiliar area and that a stranger – when asked for directions – offered to take them to their destination. Respondents are then asked which out of six presents (worth between 10 and 60 dollars) they would give to the stranger as a "thank you". Second, respondents are asked to provide a self-assessment about how willing they are to return a favor on an eleven-point Likert scale.
4. *Negative Reciprocity.* Negative reciprocity is elicited through three self-assessments. First, respondents are asked how willing they are to take revenge if they are treated very unjustly, even if doing so comes at a cost (Likert scale, 0-10). The second and

third items probe respondents about their willingness to punish someone for unfair behavior, either towards themselves or a third person.

5. *Altruism*. Altruism is measured through a combination of one qualitative and one quantitative item, both of which are related to donations. The qualitative question asks respondents how willing they would be to give to good causes without expecting anything in return on an eleven-point scale. The quantitative scenario depicts a situation in which the respondent unexpectedly receives 1,600 dollars and is asked to state how much of this amount they would donate.
6. *Trust*. The trust measure is based on one item, which asks respondents whether they assume that other people only have the best intentions (Likert scale, 0-10).

For each economic preference, the survey items are combined into a single preference measure. More specifically, each preference is computed by (i) calculating the z-scores of each survey item at the individual level and (ii) weighting these z-scores using the weights provided in Table C.1. For ease of interpretation, each preference measure is standardized to have a mean of zero and a standard deviation of one.

Table C.1: GPS Survey Items and Weights

Preference	Item description	Weight
Patience	Intertemporal choice sequence using staircase method	0.712
	Self-assessment: willingness to wait	0.288
Risk taking	Lottery choice sequence using stair case method	0.473
	Self-assessment: willingness to take risks in general	0.527
Positive reciprocity	Gift in exchange for help	0.515
	Self-assessment: willingness to return a favor	0.485
Negative reciprocity	Self-assessment: willingness to take revenge	0.374
	Self-assessment: willingness to punish unfair behavior toward self	0.313
	Self-assessment: willingness to punish unfair behavior toward others	0.313
Altruism	Donation decision	0.635
	Self-assessment: willingness to give to good causes	0.365
Trust	Self-assessment: people have only the best intentions	1

C.2 Measuring universal moral values

Moral Foundation Theory posits that people's moral concerns can be split into five foundations:

1. *Care/Harm*. This foundation measures the extent to which people care about the weak and try to keep others away from harm.
2. *Fairness/Reciprocity*. This measure captures the importance of equality, justice, rights and autonomy.
3. *In-group/Loyalty*. This foundation captures the extent to which people emphasize loyalty to the "in-group" (family, country) and how morally relevant betrayal is.
4. *Authority/Respect*. This foundation measures how important respect for authority, tradition and order is.
5. *Purity/Sanctity*. This measure captures the importance of ideas related to purity, disgust and traditional religious attitudes.

To obtain measures of the five foundations, we administer the Moral Foundations Questionnaire. In this survey, each moral foundation is measured using six different survey items. Respondents are either asked to assess the moral relevance of certain behaviors, or they are asked if they agree with certain moral value statements. All the questions are answered on a Likert scale (0–5). Table C.2 provides an overview of the specific items that are included in each foundation. In order to construct the final scores, responses are summed.

To construct a measure of the relative importance of universal versus communal moral values, we follow the approach described in Enke (2020):

$$\text{Relative importance of universal values} \quad (1)$$

$$= \text{Universal values} - \text{Communal values} \quad (2)$$

$$= \text{Harm/Care} + \text{Fairness/Reciprocity} - \text{In-group/Loyalty} - \text{Authority/Respect} \quad (3)$$

To ease interpretation, the resulting measure is standardized to have a mean of zero and a standard deviation of one.

Table C.2: Survey items: Moral Foundations Questionnaire

	Moral Relevance	Agreement with Statement
Harm/care	Emotional suffering Care for weak and vulnerable Cruelty	Compassion with suffering crucial virtue Hurt defenseless animal is the worst thing Never right to kill human being
Fairness/reciprocity	Treat people differently Act unfairly Deny rights	Laws should treat everyone fairly Justice most important requirement for society Morally wrong that rich children inherit a lot
In-group/loyalty	Show love for country Betray group Lack of loyalty	Proud of country's history Be loyal to family even if done something wrong Be team player, rather than express oneself
Authority/respect	Lack of respect for authority Conform to societal traditions Cause disorder	Children need to learn respect for authority Men and women have different roles in society Soldiers must obey even if disagree with order
Purity/sancity	Violate standards of purity Do something disgusting Act in a way that God would approve	Not do things that are disgusting Call acts wrong if unnatural Chastity is an important virtue

Note: For the items in column 1, respondents are asked to state to what extent these considerations are morally relevant (Likert scale from 0 to 5). For the items in column 2, respondents are asked to state whether they agree or disagree with the statements (Likert scale from 0 to 5).

Appendix D Pre-registration

We pre-registered the experimental design, the sampling approach, and the main analyses of our wave 2 survey at the AEA RCT Registry (#AEARCTR-0007542). This section notes deviations from the pre-analysis plan:

- Our main specification deviates from equation 1 in the pre-analysis plan by additionally including a set of control variables. As mentioned in the paper, we report results without control variables in Table A.8. The results are virtually identical.
- We did not specify how we construct the left-hand-side variable for the regressions studying treatment effects on policy support.
- We pre-registered that we would conduct additional analyses exploring heterogeneity in treatment effects by background characteristics of respondents, such as pre-treatment beliefs, political affiliation, demographic groups (i.e., age, gender, income, education) or experienced local weather. In this paper, we only report treatment effect heterogeneity by pre-treatment beliefs and political affiliation (Democrat vs. Republican) as we consider those results most relevant to the public policy debate. We note that we also find heterogeneous treatment effects with respect to some of the other background characteristics we pre-specified (e.g. women tend to react more strongly to the treatments) but that we do not detect heterogeneous treatment effects with respect to local weather conditions. Results are available upon request.
- The analyses of treatment effect heterogeneity by climate change skepticism and by moral universalism were not pre-registered, and are thus exploratory in nature.

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